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1  #!/usr/bin/env python3
2  # -*- coding: utf-8 -*-
3  """
4  Example 6.2 Decision Tree Regression
5  Machine Learning for Engineering Problem Solving
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7  """
8
9  import IPython as IP
10 IP.get_ipython().run_line_magic('reset', '-sf')
11
12 import numpy as np
13 import matplotlib.pyplot as plt
14 import sklearn as sk
15 import graphviz as graphviz
16 from sklearn.tree import export_graphviz
17
18 plt.close('all')
19
20
21 %% Train and plot a decision tree regression model
22
23 # build the data
24 m = 200
25 X = np.random.rand(m, 1)
26 y = 5 * X
27 y = y + np.random.randn(m, 1) / 10
28
29 # train the model
30 tree_reg = sk.tree.DecisionTreeRegressor(max_depth=3)
31 tree_reg.fit(X, y)
32
33 x_model = np.linspace(0, 1, 100).reshape(-1, 1)
34 y_model = tree_reg.predict(x_model)
35
36 plt.figure()
37 plt.plot(X, y, ".", label='data')
38 plt.plot(x_model, y_model, "-", label="model")
39 plt.xlabel("$x$")
40 plt.ylabel("$y$")
41 plt.legend()
42
43 %% Plot the regression tree
44
45 # create the export file for graphviz and export it. The file is exported as a
46 # .DOT file and can be viewed in an online viewer
47 https://dreampuf.github.io/GraphvizOnline/
48 export_graphviz(
49     tree_reg,
50     out_file="tree_reg",
51     rounded=True,
52     filled=True
53 )
54
55 # We can load the file back in
56 s = graphviz.Source.from_file('tree_reg')
57 s.render('tree_reg', format='jpg', view=True)
58
59
60
61

```